

Project ID :

25-26J-134

1. Topic (12 words max)

AI Powered Train Safety System with Real Time Crowd and Obstacle Detection

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The Sri Lankan railway network, while essential for daily transportation, faces critical safety and efficiency challenges. These include frequent track-related accidents (such as train-elephant collisions), inadequate guidance for passengers boarding overcrowded compartments, and limited real-time support for train drivers during hazardous conditions. This project proposes an AI-powered Train Safety and Passenger Guidance System specifically designed to address these unique issues through a combination of real-time object detection, crowd density analysis, driver decision support, and smart platform-compartment alignment.

Compartment Crowd Density Detection: In Sri Lanka, especially during peak hours, passengers struggle to board trains due to uneven crowd distribution across compartments. Overcrowded compartments pose safety hazards and reduce comfort. This system uses CCTV-based image processing, CO₂ sensors, or weight sensors to estimate crowd density within each compartment. By analyzing visual data in real-time, the system can guide platform-side signage or digital displays to inform passengers where less crowded compartments will arrive. This optimizes boarding efficiency, improves passenger safety, and reduces accidents caused by forceful boarding [3] .

Platform Compartment Alignment: Building upon the crowd density data from the compartments, this component optimizes where each train compartment should stop along the platform. Train compartments currently stop at random or varying platform sections, leading to passenger confusion, platform crowding, and delayed boarding. CCTV cameras installed along the platform monitor passenger crowd density at different segments. By integrating onboard compartment crowd data with platform crowd levels, the system calculates the ideal alignment for each compartment, directing overcrowded compartments to stop where platform crowd density is lower. This ensures a balanced distribution of passengers on the platform, facilitating faster, safer, and more efficient boarding. Platform signage and digital displays then guide waiting passengers to the appropriate boarding spots. The Compartment Crowd Density Detection data directly informs the Platform Compartment Alignment decisions. After identifying which compartments are crowded, the system aligns those compartments to less crowded platform sections, creating an optimal passenger flow from the platform onto the train [3].

Object Detection System: Sri Lanka's railway faces frequent safety risks at unmanned crossings and low-visibility track segments, where pedestrians, vehicles, or animals may unknowingly obstruct the path, especially at night or in fog. To reduce these dangers, the system uses infrared thermal sensors, heat sensors, CCTV, or sound sensors mounted on trains or at high-risk locations to detect warm-bodied obstacles in real time. These sensors identify humans, animals, or vehicles regardless of lighting conditions, calculate their size and distance, and assign a risk score. Alerts are sent to the driver or control center, and in critical cases braking is activated. For example, in elephant-prone areas like Habarana, large animals can be detected 500 - 800 meters ahead, allowing safe stopping. At unmanned crossings, early detection of crossing motorbikes helps prevent collisions. This system improves passenger safety, wildlife protection, and overall railway efficiency [1], [4], [6].

Intelligent Driver Decision Support System with Contextual Risk Assessment (Using Adaptive Rule-Based Engine and Sensor Fusion) : Sri Lanka's railway faces significant safety challenges, especially in poor visibility or unexpected track situations involving obstacles, weather, or infrastructure faults. To enhance real-time decision-making, this AI-powered Driver Decision Support System combines an adaptive rule-based decision engine with multiple sensors to continuously assess the train's environment and provide timely alerts or automatic actions. The system collects contextual data from object detection, CCTV/thermal cameras (human and animal identification) and Weather APIs (environmental conditions). A Risk Assessment Model calculates a dynamic risk score (0–100) based on factors like obstacle proximity, train speed, weather, visibility, and track curvature.

The adaptive decision engine processes this risk score by applying predefined safety rules combined with real-time sensor inputs to mimic human judgment and determine appropriate actions.

- Immediate emergency braking if a large obstacle is detected nearby.
- Alerts to slow down in conditions like “rain and curve.”
- Driver health monitoring alerts via biometric data (heart rate, drowsiness). Alerts are delivered through multiple channels: visual (dashboard messages), audio (voice commands or alarms), and haptic (seat or control vibrations). If a critical situation arises where braking distance is insufficient, the system triggers automatic braking to prevent collisions [2] .

[1] M. G. Pavlović, I. T. Ćirić, D. Ristić-Durrant, V. D. Nikolić, M. B. Simonović, M. V. Ćirić, and M. S. Banjac, "Advanced thermal camera-based system for object detection on rail tracks," in *Proc. MIPRO 2018 - 41st Int. Conv. Inf. Commun. Technol., Electron. Microelectron.**, Opatija, Croatia, 2018. [Online]. Available: [\(PDF\) Advanced thermal camera based system for object detection on rail tracks](#)

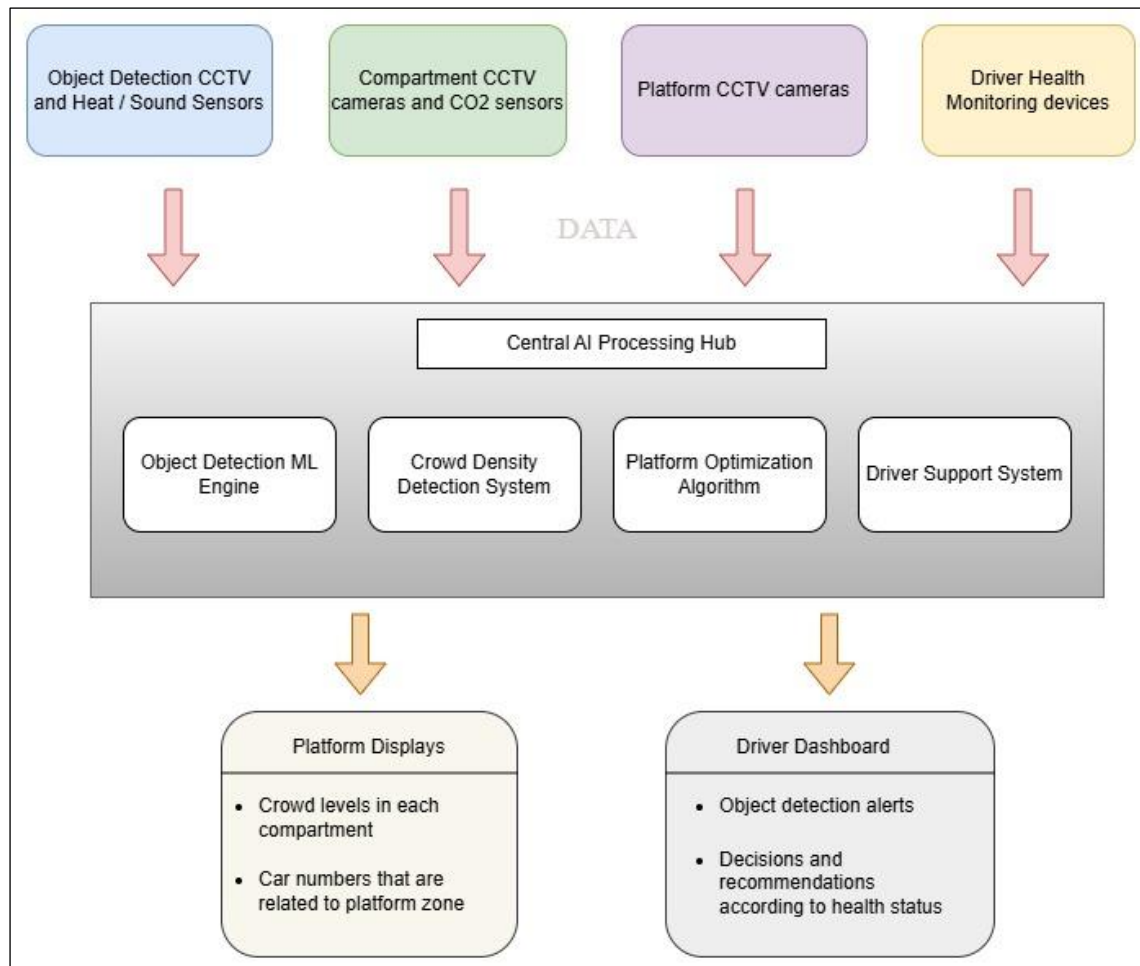
[2] S. Drakatou, K. Karakassis, and A. Dentsoras, "A fuzzy system for train driver support towards an efficient handling of engine malfunction," *ASME J. Mech. Des.*, Dec. 23, 2020. [Online]. Available: [A Fuzzy System for Train Driver Support Towards an Efficient Handling of Engine Malfunctions | IDETC-CIE | ASME Digital Collection](#)

[3] W. Helmi, R. Bridgelall, and T. Askarzadeh, "Remote sensing and machine learning for safer railways: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 168, pp. 182–203, May 2020. [Online]. Available: [Remote Sensing and Machine Learning for Safer Railways: A Review](#)

[4] South Eastern Railway (India), "SER to use AI-based system to prevent elephant-train collisions," *Indian Railways News*, Jun. 19, 2025. [Online]. Available: [SER to use AI-based system to prevent elephant-train collisions | Bhubaneswar News - Times of India](#)

[5] R. Goel, A. Sharma, and R. Kapoor, "An efficient object and railway track recognition in thermal images using deep learning," *SpringerLink*, Mar. 24, 2022. [Online]. Available: [An Efficient Object and Railway Track Recognition in Thermal Images Using Deep Learning | SpringerLink](#)

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)



This integrated system combines real-time AI analytics with multi-sensor technologies to enhance train safety and optimize passenger distribution. The solution operates through four interconnected modules that process live data streams.

The Compartment Crowd Density Detection system uses CCTV cameras combined with CO2 level monitoring devices to accurately estimate passenger occupancy in real-time. Computer vision algorithms analyze video feeds to count passengers while CO2 sensors validate occupancy levels, providing reliable crowd density measurements across all train compartments.

The Object Detection System employs CCTV cameras, heat sensors or sound sensors to detect and classify track obstacles in real-time. Machine learning algorithms process visual data, thermal signatures or acoustic patterns to identify various obstacle types, ensuring comprehensive detection capabilities across different environmental conditions.

The Platform Compartment Allocation system uses CCTV cameras positioned throughout platform areas to detect which train compartments align with specific platform zones. Computer vision algorithms track train positioning and compartment identification, enabling dynamic assignment of passenger boarding areas based on real-time crowd distribution and accessibility requirements.

The Driver Decision Support System integrates data from all modules through an intelligent scoring system that evaluates obstacles, weather conditions, and driver health status. The system monitors driver alertness and fatigue levels while providing scored risk assessments and personalized recommendations, ensuring optimal decision-making support through comprehensive health and safety monitoring.

The system architecture enables real-time data flow between components while maintaining independent operation, ensuring continuous safety monitoring and passenger guidance without single points of failure.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

1. Compartment Crowd Density Detection:

- **Specialized Domain Expertise:** Computer vision expertise for detecting and analysing passenger density using CCTV footages, CO₂ or weight sensors, and understanding passenger behaviour, especially during rush hours, for accurate crowd levels estimation.
- **Knowledge Requirements:** Deep learning models like YOLO and SSD are used to identify individuals in video streams, while sensor calibration and live data fusion techniques ensure accuracy.
- **Data Requirements and Collection Plan:** CCTV cameras and CO₂ sensors will be installed in train compartments to record passenger flow and crowd estimates, with approval from Sri Lanka Railways.

2. Object Detection System on Tracks:

- **Specialized Domain Expertise:** Deep learning and real-time object detection, particularly with CNN and YOLO, thermal, infrared imaging and familiarity with common Sri Lankan rail hazards are essential for effective obstacle detection.
- **Knowledge Requirements:** Thermal and infrared image feeds, detect heat-based anomalies, classify obstacle-related noises, and use logic to assign risk levels.
- **Data Requirements and Collection Plan:** Thermal sensors, heat sensors, CCTV, and sound sensors will be installed on trains or risky trackside areas to collect data from real or simulated obstructions for detection models.

3. Platform-Compartment Alignment System:

- **Specialized Domain Expertise:** In platform monitoring, real-time train alignment, and passenger distribution modeling.
- **Knowledge Requirements:** Vision-based crowd tracking and synchronizing train position data with platform activity.
- **Data Requirements and Collection Plan:** CCTV will track waiting passengers and train arrival alignment. Data will be merged with compartment crowd levels for optimized platform guidance.

4. Driver Decision Support System:

- **Specialized Domain Expertise:** Fuzzy logic and sensor fusion in safety-critical environments. Knowledge of biometric fatigue detection.
- **Knowledge Requirements:** Fuzzy inference systems for dynamic risk scoring and be able to design multi-modal alert mechanisms. Integration skills for working with weather APIs, and real-time track data are also essential. Additionally, knowledge in biometric sensor handling is necessary to assess driver health. Additionally, knowledge in biometric sensor handling is necessary to assess driver health.
- **Data Requirements and Collection Plan:** Biometric data from test drivers or simulations will be collected using wearable sensors or cameras. This data, combined with weather, and track condition.

8. Objectives and Novelty

Main Objective: To develop an AI-powered safety and guidance system for trains that reduces passenger congestion and detects obstacles using integrated simulation, IoT sensors, computer vision, and AI-driven decision support mechanisms to enhance safety, efficiency, and commuter experience.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22283412 Udaya Sri A L S	Implement real-time compartment crowd detection	Develop a crowd estimation system using either computer vision or CO ₂ , or weight sensors based on availability	This system uniquely combines advanced computer vision techniques with real-time passenger guidance to directly influence boarding behavior, addressing the challenge of uneven crowd distribution in Sri Lankan trains. Unlike traditional methods that passively monitor crowd density, our approach proactively uses dynamic platform-side signaling informed by accurate, real-time compartment crowd estimates—even under complex conditions like occlusions and poor lighting. This integration of real-time analytics with actionable passenger direction is rarely implemented in existing railway systems, making our solution distinctively effective in enhancing both safety and operational efficiency.

IT22250988 Bandara I K	Optimize compartment-to-platform alignment	Use platform CCTV feeds to estimate platform crowd and adjust train stops accordingly	Existing smart station systems typically use pre-mapped boarding zones or multiple sensors on platforms. Our system uniquely depends on real-time video from CCTV to guide both train alignment and passenger movement. It reverses the traditional model by using compartment crowd data to decide where the train should stop, not just where passengers should wait. This data-driven alignment is novel for countries with informal or unmapped platforms.
IT22291578 Prarthana L D R	Develop an obstacle detection system on tracks	Use a single sensing method (either thermal sensor, CCTV, sound sensor, or heat sensor) per deployment to detect obstacles on or near the railway track. Process real-time input to identify humans, animals, or vehicles. Calculate distance and size to assign a risk score. Trigger alerts.	Object detection system uniquely enhances safety by dynamically calibrating thermal sensors to tropical conditions, reducing false alarms from environmental noise. It employs real-time risk scoring that evaluates obstacle size, distance, and movement to predict collision likelihood, enabling timely alerts. The system uses geofencing to adjust sensitivity in high-risk areas minimizing false alerts while maximizing safety. Optimized for low-power edge devices, it functions reliably in remote regions without constant network access, addressing infrastructure challenges often ignored by other systems.

IT22597342 Jayasinghe J M H C	Design driver decision support system	Develop fuzzy logic rules for decision-making; integrate driver health inputs	Context-aware decision-making framework that closely mimics human judgment through a transparent rule-based engine, enabling fast, explainable & reliable responses to complex and dynamic railway conditions. Unlike conventional black-box AI models, this approach integrates diverse environmental and biometric inputs into a unified risk assessment that dynamically adjusts recommendations based on real-time context such as weather, track curvature & driver alertness. Its multi-channel alert system combined with automatic intervention capabilities ensures proactive safety management, effectively bridging the gap between sensor data and actionable driver support in challenging operational environments.
-------------------------------------	---------------------------------------	---	---

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22283412 Udaya Sri A L S	Applies computer vision algorithms and machine learning models to process sensor data streams, enabling adaptive passenger counting through statistical modeling and real-time analytics for dynamic crowd estimation.
IT22250988 Bandara I K	Implements computer vision analytics and predictive algorithms on CCTV feeds to estimate crowd density, analyze historical patterns, and optimize train positioning through machine learning models.
IT22291578 Prarthana L D R	The obstacle detection system uses infrared thermal sensors, heat sensors, CCTV, or sound sensors mounted on trains or at high-risk locations to detect warm-bodied obstacles in real time. It calculates size, distance, and risk scores, triggering alerts and automatic braking to enhance safety at unmanned crossings and low-visibility areas.
IT22597342 Jayasinghe J M H C	Integrates fuzzy logic algorithms with biometric data processing, applying decision tree models and predictive analytics to combine environmental sensors with physiological monitoring for comprehensive support.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mrs.	Lakmini	Abeywardhana	
Co-Supervisor	Mr.	Amila	Alexander	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
-----	--	----	--

- b) Does the proposed topic exhibit novelty?

Yes		No	
-----	--	----	--

- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
-----	--	----	--

- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
-----	--	----	--

- e) Supervisor's Evaluation and Recommendation for the Research topic:

--

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.